

Task 4: Continuous Integration (CI) – devops-crm-project

1. Objective

The objective was to implement and verify Continuous Integration (CI) for the devops-crm-project using GitHub Actions. The CI process automatically installs dependencies, lints the code, runs type checking and tests, and builds the application for Pull Requests.

2. Existing Workflows Found

Before making changes, I checked the `.github/workflows/` directory and found:

- `ci.yml` – existing Continuous Integration workflow
- `cd.yml` – Continuous Deployment workflow
- `publish.yml` – workflow that publishes the app to npm

3. Existing CI Workflow

The workflow runs on pushes to main and on Pull Request events. It also uses read-only contents permission and concurrency settings that cancel older CI runs when a newer commit is pushed.

The workflow creates a temporary Twenty CRM test instance using the Twenty GitHub Action. This is required because the integration tests need a live Twenty server. The server URL and API key are passed to the tests through `TWENTY_API_URL` and `TWENTY_API_KEY`.

4. Existing CI Steps

1. Install dependencies using `yarn install --immutable`.
2. Run linting using `yarn lint`.
3. Run type checking using `yarn typecheck`.
4. Run unit tests using `yarn test:unit`.
5. Run integration tests using `yarn test` with the temporary Twenty server.

5. Missing Requirement

The existing CI workflow did not have an explicit application build step. The task required the application to be built during CI, so this was the gap I needed to fix.

6. Change Made

I added this step at the end of the existing CI job:

Build application

```
yarn twenty dev:build
```

This Twenty CLI command builds the application and writes output to `.twenty/output`. I also tested it locally with the Twenty Docker server stopped and confirmed that it works independently.

No other parts of `ci.yml` were changed.

7. Steps I Followed

1. Explored `.github/workflows/` and found `ci.yml`, `cd.yml`, and `publish.yml`.
2. Read the existing `ci.yml` before editing it.
3. Confirmed via `yarn twenty --help` that `dev:build` is the correct build command.
4. Tested `yarn twenty dev:build` locally with the Twenty Docker server stopped.
5. Created the `fiza-task4` branch from `main`.
6. Added only the build step to `ci.yml`.
7. Used `git diff` to confirm the change was minimal.
8. Committed and pushed the branch.
9. Opened a Pull Request from `fiza-task4` to `main`.
10. Verified that GitHub Actions ran automatically and all steps passed.

8. Git and Pull Request Workflow

```
git checkout main
```

```
git pull origin main
```

```
git checkout -b fiza-task4
```

```
git diff .github/workflows/ci.yml
```

```
git add .github/workflows/ci.yml
```

```
git commit -m "Add build step to existing CI workflow"
```

```
git push -u origin fiza-task4
```

Pull Request: fiza-task4 → main in PearlThoughts-Intern-DevOps/devops-crm-project.

9. CI Verification

- Opened the Pull Request on GitHub.
- Checked the Actions tab and Pull Request checks.
- Confirmed the workflow started automatically.
- Confirmed checkout, Twenty test instance, dependency installation, lint, typecheck, unit tests, integration tests, and the new build step all passed.

10. Final Result

The CI workflow now automatically validates Pull Requests and includes the required application build step. The existing CI setup was preserved, the new build command was verified locally, and the GitHub Actions workflow passed successfully.